HW11

Crosstalk simulation of eight coupled channels

Anonymous technical report | Reproducible frequency-to-time-domain analysis

Scope and result

The supplied 16-port network is terminated by the specified source and load RC networks. A completed Python solver computes all transmitter-to-receiver transfer functions, then combines eight independently generated NRZ signals using linear convolution. The results below use the regenerated run saved with this package, rather than the example images in the assignment.

Empirical voltage-cluster quantile gap	Rx0	Rx4
Only corresponding transmitter on	0.286 V	0.2945 V
All eight transmitters on	0.2045 V	0.157 V
Decrease from isolated case	0.0815 V	0.1375 V

This diagnostic summarizes empirical voltage-cluster separation over a finite deterministic record. It is not a bit-conditioned or worst-case eye opening, bit-error-rate estimate, or receiver compliance limit. The exact definition and numerical checks are given on the following pages.

Deliverables

The report includes all sixteen S_{1j} curves, the isolated and all-on eyes at Rx0 and Rx4, the requested 4 ns Rx1-Rx4 waveform snapshot with only Tx4 active, and all eight all-on receiver eyes. The complete source, saved numerical summaries, and validation tests are supplied alongside this document.

Optional ADS comparison

ADS was not run for this report. The supplied assignment includes instructor example ADS images; no numerical agreement with a fresh ADS simulation is claimed here.

1 | Problem definition

Physical Touchstone ports are numbered 1-16. The program's channel labels are zero-based: Tx k drives physical port 2k+1 and Rx k observes physical port 2k+2, for k = 0,...,7. Thus Tx4 is physical port 9 and Rx4 is physical port 10.

Parameter	Specified / simulated value
Signaling	10 Gbit/s single-ended NRZ; UI = 100 ps; source voltage levels 0 and 1 V
Rise and fall transitions	20 ps linear transitions
Source termination	R_S = 30 ohm; C_S = 1.5 pF shunt at each Tx port
Receiver termination	R_L = 65 ohm; C_L = 1 pF in parallel at each Rx port
Initial source delays	Tx0, Tx2, Tx4, Tx6: 0 ps; Tx1, Tx3, Tx5, Tx7: 50 ps (0.5 UI)
Input data	16-port S network; 4,001 frequency samples; 10 MHz spacing; maximum 40 GHz
Reference impedance	Common real Z_0 = 50 ohm
Retained record	262,144 bits per transmitter after the steady-state guard

Required-output map

Assignment item	Location in this report
1. S11, S12, ..., S1,16	Section 4
2-3. Rx0 with only Tx0 / all Tx on	Section 5, top row
4-5. Rx4 with only Tx4 / all Tx on	Section 5, bottom row
6. Rx1-Rx4 snapshot with only Tx4 on	Section 6
Compute the eyes at every Rx	Appendix A, all eight all-on eyes

Inactive transmitters

Turning a transmitter off sets its source waveform to zero while retaining its specified source resistor and shunt capacitor. Removing inactive terminations would change the circuit and would not be the isolated-transmitter comparison used here.

2 | Multiport circuit model

For each frequency, voltage-wave scattering uses $v^- = S\, v^+$. With port currents defined into the S-parameter network and a common real reference impedance, $v = v^+ + v^-$ and $Z_0 i = v^+ - v^-$. Eliminating the wave amplitudes gives:

```
(I - S) v - Z0 (I + S) i = 0
```

A source v_{Tj} in series with R_S is replaced by a Norton current v_{Tj}/R_S in parallel with R_S. The external port admittance is G + j omega C, with diagonal entries $1/R_S$ and C_S at Tx ports, and $1/R_L$ and C_L at Rx ports. KCL and the scattering relation form the 32-by-32 system:

```
[ G+j*omega*C       I             ] [v]   [B]
[ I-S          -Z0*(I+S)          ] [i] = [0]
```

The eight columns of B excite one transmitter each. Column j has $1/R_S$ at Tx j's physical port and zero elsewhere, so the voltage part of the solution is already normalized to a 1 V Thevenin source. Selecting the eight Rx rows yields $H_{rj}(f) = V_{Rx\,r}(f) / V_{Tj}(f)$. All right-hand sides are solved together for each frequency; a matrix inverse is not explicitly formed.

```
y_r(t) = sum_j [ h_rj(t) * x_j(t) ]
```

The network and external RC terminations are linear and time invariant. Therefore one may compute each transmitter's response at all receivers and add them. The stored transfer-function axes are frequency, receiver, transmitter; selecting a transmitter means selecting the third axis.

Why S/2 alone is insufficient here

With all ports resistively matched to Z_0 and no extra capacitors, the off-diagonal source-to-load transfer is $S_{ij}/2$. This useful normalization check is not the actual HW11 circuit: the 30 ohm / 65 ohm terminations and shunt capacitors require the loaded multiport solve above.

Coupling interpretation

A diagonal channel H_{rr} contains its loaded transmission response. Off-diagonal H_{rj} terms are crosstalk paths. Nearby aggressors contribute the largest interference in this data set; their 0.5 UI source offset places many transitions near another channel's decision region. The channel's own intersymbol interference and the added aggressor terms must both be retained.

3 | Frequency-to-time method

The supplied S-parameter matrix is read without assuming reciprocity and checked for uniform frequency spacing. The loaded H spectrum is multiplied by the Kaiser window below over the supplied 0-40 GHz band, then zero-padded to the 250 GHz sampling Nyquist frequency. DC is constrained to be real. Hermitian negative-frequency symmetry and a 50,000-point real inverse DFT produce dimensionless discrete-convolution weights.

```
W(f) = I0(pi*a*sqrt(1-(f/fmax)^2)) / I0(pi*a)
```

Numerical setting	Saved run
Samples per UI	50
Time step	2 ps
Kaiser frequency-window parameter	1.0
Retained negative impulse time	5 ns
Retained positive impulse time	20 ns
Steady-state guard / convolution section	1000 UI / 32,768 samples
NRZ random streams	Separate deterministic SeedSequence child streams, one per Tx
Master seed	2026

A finite frequency bandwidth produces pre-ringing and long impulse tails. The solver retains the specified negative- and positive-time support rather than discarding all negative lags or circularly wrapping the response into the data. Input bits are generated for a guard on both sides of the retained record. Finite 20 ps linear edges and the alternating 50 ps source delays are applied to each transmitter before convolution.

Sectional convolution

Linear convolution is evaluated with FFT overlap-add. Section responses are added at their correct absolute offsets; padding prevents within-section circular aliasing. Only the guarded central steady-state interval is used for eyes and metrics. Separate child streams avoid the accidental identical-pattern correlation that would result from resetting every transmitter to one seed.

Eye and voltage-cluster diagnostic

Each two-UI eye is a histogram of voltage samples folded by the fixed UI. Each phase column is normalized to observed bin mass; the color scale is $\log_{10}$(probability per voltage bin), not probability density or BER. The eye phase is absolute sample time modulo 100 ps; the data are not recentered or aligned to recovered bits.

The saved `opening_v` diagnostic splits voltage samples at half the main-path DC gain (about 0.340664 V). At each of 50 phases, it subtracts the lower cluster's 99.9% quantile from the upper cluster's 0.1% quantile, requiring at least 20% of samples in each cluster. The largest gap is reported. The histogram has 3,000 bins from -0.25 to 1.25 V, giving 0.5 mV voltage resolution. This voltage-conditioned diagnostic is not bit-conditioned and cannot establish a target-BER eye opening or detect all possible error trajectories.

The bandwidth, window, impulse support, sampling, and finite record define this approximation. No random jitter, receiver noise, clock recovery, or equalizer optimization is included.

4 | S-parameter responses

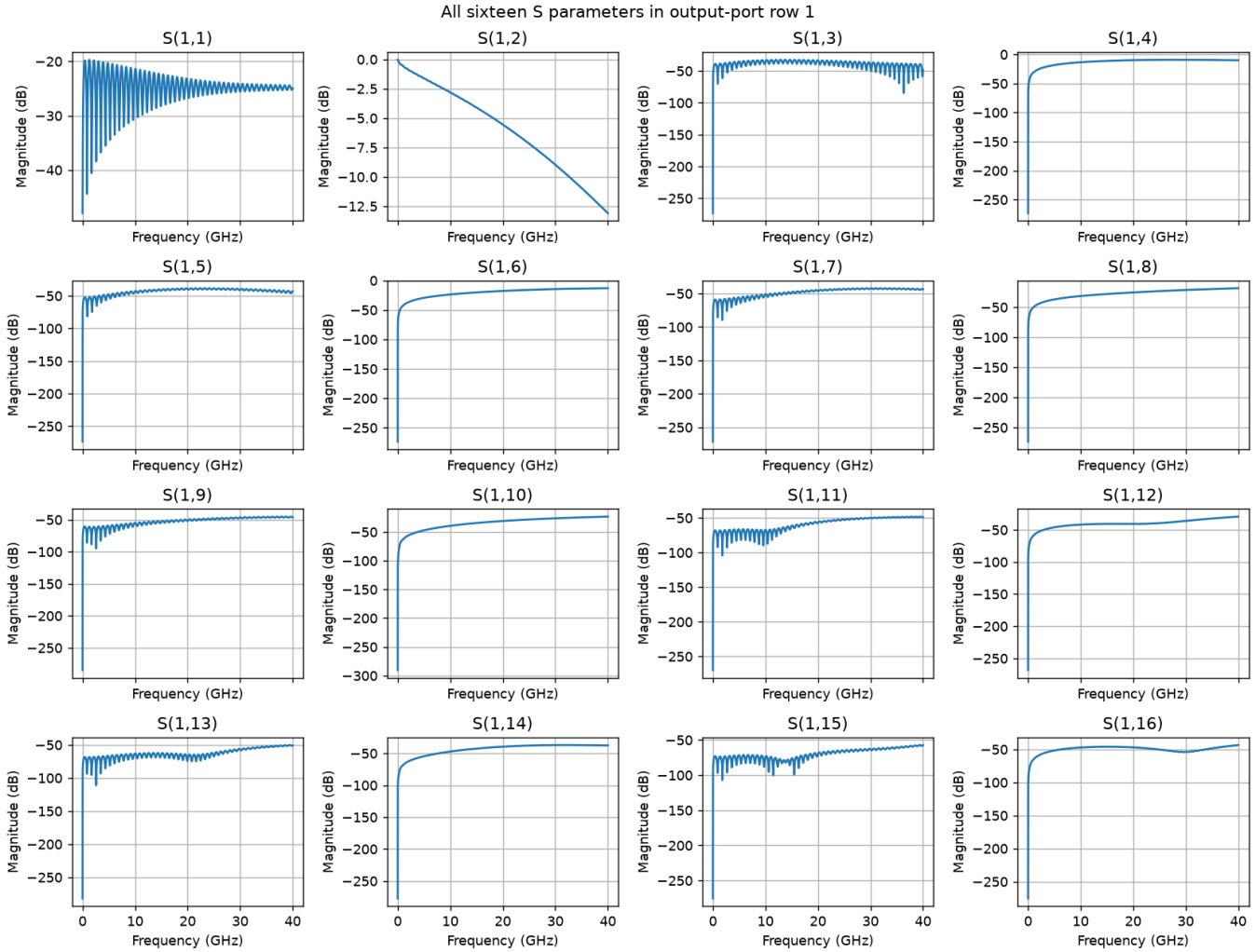


Figure 1. Regenerated magnitude responses for all sixteen S_{1j} elements. Indices here are the original one-based physical Touchstone port indices. These curves belong to the bare network, before the external source and load RC networks are added.

The S_{1j} request is a row of the multiport matrix. It must not be confused with the eight loaded source-to-Rx responses. In particular, S_{12} connects the two ends of channel 0, while S_{11} is the reflection at physical port 1. The simulation uses the full matrix throughout, including all coupling and reflections.

The physical channel port pairs are (1,2), (3,4), (5,6), (7,8), (9,10), (11,12), (13,14), and (15,16). Pairing adjacent entries in a block of odd-numbered ports would therefore be incorrect.

5 | Required receiver-eye comparisons

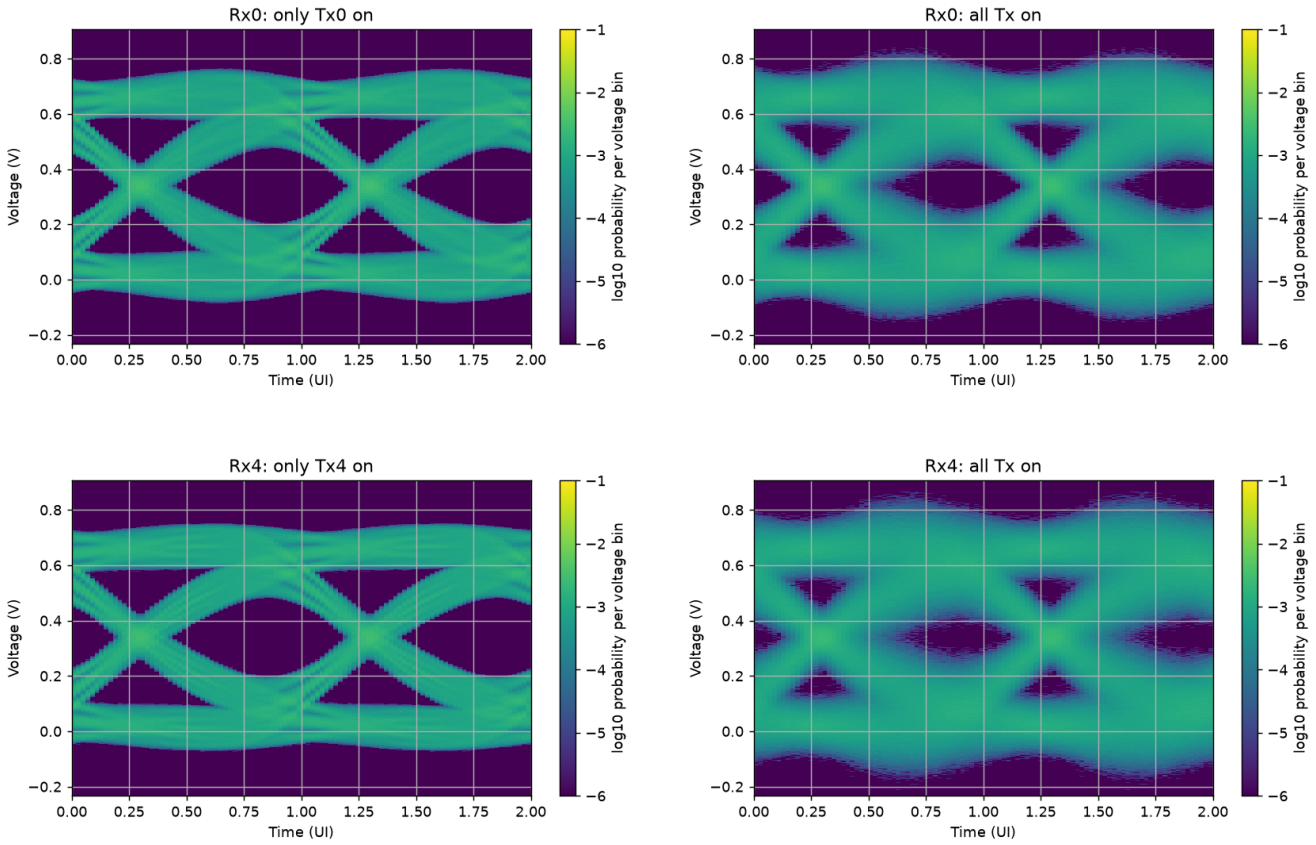


Figure 2. Top: Rx0 with only Tx0 active (left) and all transmitters active (right). Bottom: Rx4 with only Tx4 active (left) and all transmitters active (right). All plots are regenerated from this run with the same numerical settings and transmitter data reused between each isolated/all-on comparison.

The source voltage levels remain 0 and 1 V; the receiver levels differ because of loaded transmission loss and reflections. Adding aggressors increases the range of trajectories at a given phase and reduces the voltage-cluster gap. Rx4 lies within the group of eight channels, so coupling arrives from both sides; Rx0 is an edge channel.

The display represents finite sample occupancy. Unoccupied histogram bins mean that the simulated record did not visit those bins. This does not establish a zero error probability.

6 | Tx4-only waveform snapshot

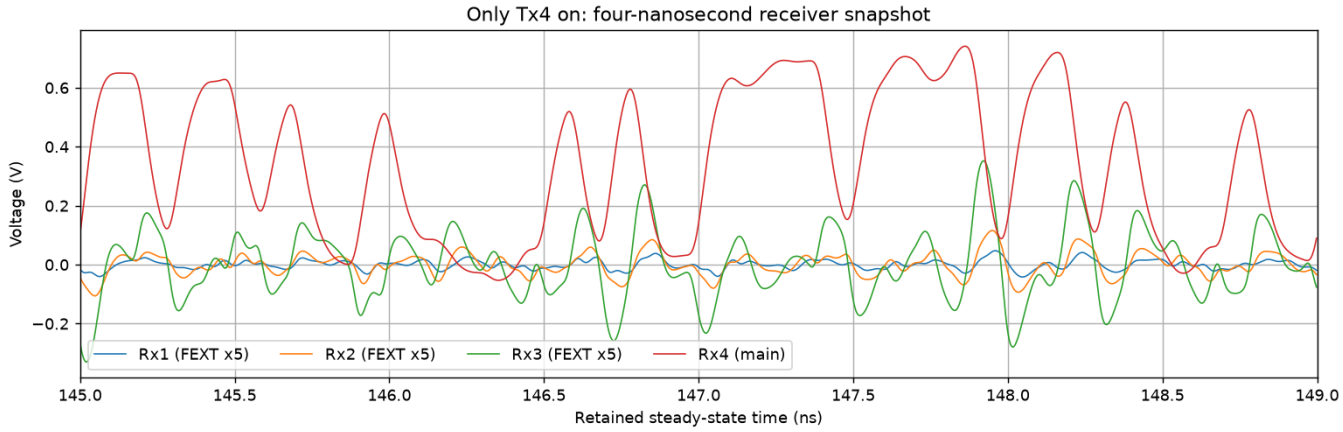


Figure 3. Requested 4 ns snapshot at Rx1, Rx2, Rx3, and Rx4 with only Tx4 active. The plot explicitly identifies any 5x display scaling of Rx1-Rx3 crosstalk; Rx4 is the directly driven channel. Display scaling is not part of the simulation or eye metrics.

The response at Rx4 carries the main NRZ pattern. The responses at the other receivers are caused by coupling from Tx4, with no independently driven data on those receivers' corresponding transmitters. They show transition-sensitive pulses and ringing rather than a separate NRZ data stream.

Receiver labels are zero-based. Rx1, Rx2, Rx3, and Rx4 correspond to physical ports 4, 6, 8, and 10, respectively; the active source Tx4 is physical port 9. The closest displayed coupled receiver is Rx3.

The snapshot is taken from the guarded steady-state output, not the startup transient. Independent random patterns are required for the all-on test; this isolated snapshot intentionally has only one nonzero source waveform.

Appendix A | All eight receiver eyes

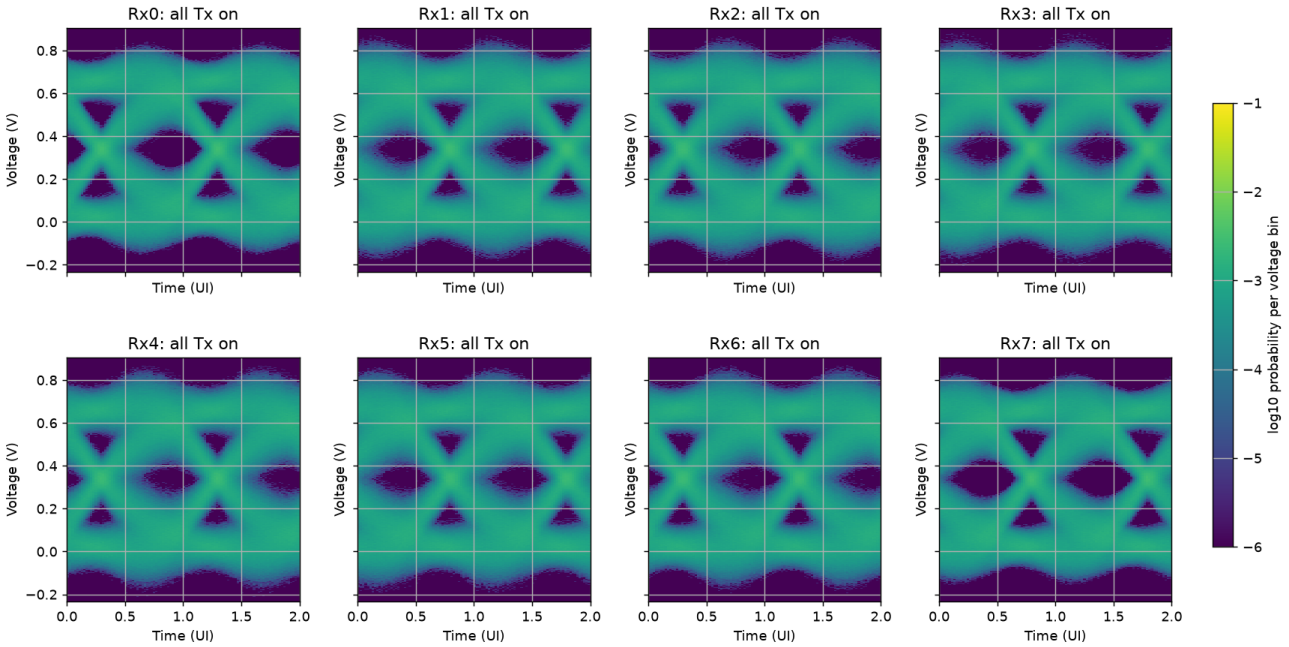


Figure 4. All eight all-on eyes computed from the same retained record and the same absolute time-modulo-UI reference. The appendix satisfies the requirement to compute all receiver eyes, including those not highlighted in the main comparisons.

Receiver	Physical Rx port	Cluster gap (V)	Best phase (UI)
Rx0	2	0.2045	0.88
Rx1	4	0.16	0.38
Rx2	6	0.1565	0.88
Rx3	8	0.161	0.38
Rx4	10	0.157	0.9
Rx5	12	0.1585	0.38
Rx6	14	0.154	0.9
Rx7	16	0.203	0.38

Appendix B | Numerical verification

The solver is accompanied by independent circuit and convolution checks. The saved verification summary records the numerical outcomes; tests exercise matched-load normalization, independent circuit equations, port ordering, source independence, linear superposition, and finite convolution alignment. Truncation diagnostics and a larger-support short-record comparison quantify numerical approximation rather than claiming an exact infinite-bandwidth waveform.

Recorded numerical diagnostics

Quantity	Measured value
DFT length / frequency-grid period	50,000 / 100 ns
Maximum discarded impulse L1 norm, one path	0.001004526
Maximum discarded impulse L2 norm, one path	8.566122e-06
Maximum retained-tap DC-sum error	0.9850817 mV per 1 V source
Negative-lag energy fraction	1.933581e-07
Larger support for comparison	-10 ns to +40 ns
Comparison record, all eight Tx active	2,048 bits per transmitter
Maximum waveform difference, larger support	0.3258419 mV
RMS waveform difference, larger support	0.2438926 mV
Sampling comparison, all eight Tx active	50 vs 100 samples/UI; 512 bits
Maximum / RMS sampling difference	0.1100771 / 0.02882472 mV
Simulation arithmetic	Float32 FFT and accumulation; float64 circuit/diagnostics

The larger-support comparison checks this particular short all-on sequence; it is not a universal worst-case bound. The residual DC-sum error explicitly records a small support-truncation bias. Kernel samples are dimensionless discrete-convolution weights, so no additional time-step multiplier is applied.

Independent checks: passed

All 16 independent unit tests passed with 0 failures and 0 errors. Saved-artifact verification additionally checked all 4,001 frequencies and 64 loaded transfer paths.

Independent comparison	Maximum absolute error
MNA vs wave-boundary circuit equations	5.850233e-15
Saved impulse vs direct trigonometric sum	4.594261e-10
Float64 overlap-add vs direct convolution	5.329071e-15
Saved Tx4 snapshot vs direct convolution	0.2451182 microV
Maximum off-diagonal source-bit correlation	0.005073805

Appendix C | Reproduction and source record

Audit scope

Every all-on receiver histogram conserves 13,107,200 samples, or 262,144 samples per phase, with no samples outside the histogram voltage range. Ten desired-channel percentile gaps were independently recalculated. The entire full-record waveform is not stored or independently reconstructed; the direct waveform check covers the saved snapshot.

Reproduce the complete saved run

Use Python 3.12 or newer and install `requirements.txt`. In the HW11 directory, run `python verify_inputs.py`, `python analyze.py`, `python tests/verify_results.py`, then `python report/build_report.py`. The artifact audit includes the unit tests. The report builder requires matching hashes for the analysis source and all three saved numerical artifacts.

The explicit analysis command, run from the repository root, is shown wrapped below; enter it as one line:

```
python homework/hw11/analyze.py --bits 262144 --seed 2026
  --samples-per-ui 50 --block-size 32768
  --kaiser-a 1 --pre-ns 5 --post-ns 20
```

Recorded environment	Version
Python	3.14.4
NumPy	2.5.0
SciPy	1.18.1
Matplotlib	3.11.2

The master seed and separate child-stream construction are fixed. The software versions are recorded because exact pseudorandom and floating-point reproduction can depend on the numerical environment. The source hash below binds the saved results to the analysis code.

Source record

The authoritative assignment is `instructions/hw11_assignment.pdf`. The supplied template, clarification notes, network data, and legacy support files are retained as references. The manifest records original and staged hashes. The incomplete template and obsolete manual same-seed advice are not the executable workflow. The prior personal student report is excluded.

Dataset SHA-256: 07d8ebd6daebc3075ec5a4bb9ea9996c530ea2bdd3b42858e0b2b4cd172e2cdb

Analysis SHA-256: 200a311fd6e9b411f9625dcb4fbb9f36f946dadd5c8504967d719b3b909cd042